Uniform critical endpoints of asymptotic Borel surjectivity

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Abstract

Both uniform critical endpoints are excluded for every strongly regular sequence. The nonuniform critical endpoint inclusion remains unresolved here.

Catalogue scope. AIM 708; partial progress. Canonical target at revision `1e56463`.

1 Statement and argument

This section proves two of the three assertions in Problem 708:

$$\mathcal{S}(\mathcal{A}_M) = \mathcal{S}(\widetilde{\mathcal{A}}^u_M) = (0, \gamma(M)) \tag{1}$$

for every strongly regular sequence M. The nonuniform endpoint inclusion $\gamma(M) \in \mathcal{S}(\widetilde{\mathcal{A}}_M)$ is not proved here. Thus the complete catalogue target is not counted as resolved.

Here $M = (M_p)_{p\geq 0}$, $M_0 = 1$, is strongly regular if it is positive and logarithmically convex, and there are $A_0, B_0 > 0$ such that

$$M_{p+q} \leq A_0^{p+q} M_p M_q, \qquad \sum_{q\geq p} \frac{M_q}{(q+1)M_{q+1}} \leq B_0 \frac{M_p}{M_{p+1}}.$$

Writing $m_p = M_{p+1}/M_p$, define

$$\gamma(M) = \sup\{a > 0 : (m_p/(p+1)^a)_p \text{ is almost increasing}\}.$$

Almost increasing means $c_p \leq K c_q$ for $p \leq q$ and one fixed K. The sector S_α is the unbounded sector $|\arg z| < \pi\alpha/2$ on the logarithm's Riemann surface. An M-bounded series $\sum a_p z^p$ satisfies $|a_p| \leq C A^p M_p$ for some $C, A > 0$. The derivative class $\mathcal{A}_M(S_\alpha)$ satisfies $\sup_{S_\alpha} |f^{(p)}| \leq C A^p p! M_p$. The class $\widetilde{\mathcal{A}}^u_M(S_\alpha)$ consists of holomorphic functions for which one pair C, A satisfies

$$\left| f(z) - \sum_{p<N} a_p z^p \right| \leq C A^N M_N |z|^N \qquad (z \in S_\alpha,\ N \geq 0). \tag{2}$$

The nonuniform class permits constants depending on each bounded proper subsector. The Borel map sends a function to its unique formal series, and $\mathcal{S}$ denotes the openings where it is onto all M-bounded series.

We use the established rational-opening criterion [1, Theorem 4.17 and Corollary 4.18]: if N is strongly regular and $r > 0$ is rational, then uniform-asymptotic surjectivity on S_r holds exactly when $r < \gamma(N)$. The following inverse-Borel transform transfers a hypothetical critical endpoint to a smaller rational critical endpoint without losing an angular margin.

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

Theorem 1.1 (Inverse-Borel transfer)**.** *Let M be strongly regular and assume that the Borel map from $\widetilde{\mathcal{A}}^u_M(S_\alpha)$ is surjective. If $0 < s < \alpha$ and $Q_p = M_p/\Gamma(1+sp)$, then the Borel map from $\widetilde{\mathcal{A}}^u_Q(S_{\alpha-s})$ onto all Q-bounded formal series is surjective.*

Proof. **A decaying cutoff preserving every coefficient.** Surjectivity supplies a function F whose formal series is z. It is bounded because the estimate with $N = 0$ is global. Put $U(z) = F(z)/z$. The estimate (2) at order $N + 1$ and moderate growth give

$$|U(z) - 1| \le CA^{N+1}M_{N+1}|z|^N \le C'B^N M_N|z|^N \qquad (N \ge 1).$$

The estimate for F at order 1 bounds U near the vertex, and boundedness of F bounds U at infinity. Thus $U \in \widetilde{\mathcal{A}}^u_M(S_\alpha)$ has formal series 1 and

$$|U(z)| \le C \min\{1, |z|^{-1}\}. \tag{3}$$

If f realizes any prescribed series, multiplication by U preserves every coefficient and retains its uniform M-asymptotics: for $P_{N-1}(z) = \sum_{p<N} a_p z^p$,

$$Uf - P_{N-1} = (f - P_{N-1}) + (U - 1)f,$$

and f is bounded while $U - 1$ is uniformly M-flat. Repeating this operation, we may arrange, for any fixed integer $L \ge 1$,

$$|f(z)| \le C_L \min\{1, |z|^{-L}\}. \tag{4}$$

There is no change of opening and no pole in this cutoff construction.

The inverse-Borel contour. For $z \in S_{\alpha-s}$, choose $\vartheta \in (\pi/2, 3\pi/4)$ sufficiently close to $\pi/2$ that both rays $\arg t = \arg z - s\vartheta$ and $\arg t = \arg z + s\vartheta$ lie inside S_α. Let H_R travel from the vertex outward along the lower ray to radius R, counterclockwise along the circular arc to the upper ray, and back to the vertex along that ray. Define

$$(\mathcal{B}_s f)(z) = \frac{1}{2\pi i s} \int_{H_R} e^{(z/t)^{1/s}} f(t)\, \frac{dt}{t}. \tag{5}$$

All powers are taken on the logarithm surface. On the rays the real part of $(z/t)^{1/s}$ is strictly negative, making the integral convergent at the vertex. Cauchy deformation proves independence of R and of nearby admissible ray choices: the connecting small arcs tend to zero exponentially. The local definitions glue to a holomorphic function on the exact sector $S_{\alpha-s}$. For holomorphy fix a contour on a neighborhood of any chosen z; independence allows its radius and rays to be chosen subsequently for estimates.

Under $v = (z/t)^{1/s}$, the contour becomes the usual Hankel contour with rays in the left half-plane. The reversed orientation cancels the minus sign in $dt/t = -s\, dv/v$. The reciprocal-gamma Hankel formula therefore gives

$$\mathcal{B}_s(t^p)(z) = \frac{z^p}{\Gamma(1 + sp)} \qquad (p \ge 0). \tag{6}$$

In particular $\mathcal{B}_s 1 = 1$.

Uniform remainders up to the angular edges. Put $R_N(t) = f(t) - \sum_{p<N} a_p t^p$. Its bound is

$$|R_N(t)| \le CA^N M_N |t|^N \qquad (t \in S_\alpha).$$

For $N \ge 1$, use the contour radius $R = |z|/(sN)^s$. On either ray the exponential has modulus at most 1, so the absolute contribution of R_N is bounded by a fixed multiple of

$$CA^N M_N \int_0^R r^{N-1}\, dr = CA^N M_N \frac{R^N}{N}.$$

The circular arc has angular length at most $3\pi s/2$, and its kernel has modulus at most e^{sN}. Hence

$$\left|\mathcal{B}_s f(z) - \sum_{p<N} \frac{a_p}{\Gamma(1+sp)} z^p\right| \le C_s A^N M_N |z|^N (sN)^{-sN} \left(e^{sN} + \frac{1}{N}\right).$$

Stirling's estimate, with its polynomial factor absorbed into an exponential constant, yields

$$\left|\mathcal{B}_s f(z) - \sum_{p<N} \frac{a_p}{\Gamma(1+sp)} z^p\right| \le C B^N \frac{M_N}{\Gamma(1+sN)} |z|^N = C B^N Q_N |z|^N \qquad (N \ge 1). \tag{7}$$

The constants are independent of the angle of z throughout the full output sector. No inverse angular cosine appears.

Boundedness on the unbounded output sector. The estimate at $N = 1$ bounds $\mathcal{B}_s f$ for $|z| \le 1$. For $|z| \ge 1$, choose $R = |z|$ in (5). The arc contribution is bounded because the kernel has modulus at most e and f is bounded. On a ray of angle θ, write

$$f(re^{i\theta}) = a_0 \mathbf{1}_{(0,1]}(r) + k_\theta(r).$$

The first remainder estimate near the vertex and (4) at infinity give

$$\sup_\theta \int_0^\infty |k_\theta(r)| \frac{dr}{r} < \infty. \tag{8}$$

Thus k_θ contributes a uniformly bounded amount on each ray. For the constant term put $\eta = \exp(i(\arg z - \theta)/s)$, so $|\eta| = 1$ and $\operatorname{Re}\eta < 0$. Substitution gives

$$\int_0^1 e^{\eta(|z|/r)^{1/s}} \frac{dr}{r} = s \int_{|z|^{1/s}}^\infty e^{\eta y} \frac{dy}{y}.$$

Integration by parts bounds the last expression by $2s/|z|^{1/s}$, uniformly even as $\operatorname{Re}\eta \uparrow 0$. This proves boundedness on $S_{\alpha-s}$ and supplies $N = 0$ in (7).

Given any Q-bounded coefficients b_p, set $a_p = b_p \Gamma(1+sp)$. These are M-bounded. Realize them by the assumed surjectivity, apply the coefficient-preserving cutoff, and then (5). Its coefficients are b_p by (6), and (7), including $N = 0$, proves the transfer theorem. □

Lemma 1.2 (Regularizing the divided sequence)**.** *If $0 < s < \gamma(M)$, then $Q_p = M_p/\Gamma(1+sp)$ is exponentially equivalent to a strongly regular sequence Q' satisfying $\gamma(Q') = \gamma(M) - s$. Here exponential equivalence means $C^{-p} Q_p \le Q'_p \le C^p Q_p$ for a fixed $C \ge 1$; it preserves both the formal-series class and the uniform-asymptotic class.*

Proof. Put

$$\ell_p = \frac{\Gamma(1+s(p+1))}{\Gamma(1+sp)}, \qquad q_p = \frac{Q_{p+1}}{Q_p} = \frac{m_p}{\ell_p}.$$

The gamma quotient satisfies $\ell_p \asymp (p+1)^s$. Since $s < \gamma(M)$, (q_p) is almost increasing. Define

$$\widetilde{q}_p = \inf_{j \ge p} q_j, \qquad Q'_0 = 1, \qquad Q'_p = \prod_{j=0}^{p-1} \widetilde{q}_j.$$

For some $K \ge 1$, $K^{-1} q_p \le \widetilde{q}_p \le q_p$. Thus Q' is logarithmically convex and $K^{-p} Q_p \le Q'_p \le Q_p$.

Moderate growth holds for Q itself: the logarithmically convex sequence $D_p = \Gamma(1+sp)$ has $D_0 = 1$ and $D_{p+q} \ge D_p D_q$, so

$$Q_{p+q} \le A_0^{p+q} Q_p Q_q.$$

Exponential equivalence transfers moderate growth to Q'.

Choose $s<\beta<\gamma(M)$. Almost increase of $m_p/(p+1)^\beta$ and comparison of the gamma quotient imply, for $j\ge p$,

$$\widetilde{q}_j \ge c\,\widetilde{q}_p\left(\frac{j+1}{p+1}\right)^{\beta-s}.$$

Because $\beta-s>0$, summation gives

$$\sum_{j\ge p}\frac{1}{(j+1)\widetilde{q}_j}\le\frac{C}{\widetilde{q}_p},$$

which is strong nonquasianalyticity. Finally $\widetilde{q}_p\asymp m_p/(p+1)^s$, so the defining almost-increase conditions give exactly $\gamma(Q')=\gamma(M)-s$. □

Theorem 1.3. *For every strongly regular sequence M, the equality (1) holds.*

Proof. The known interior and exterior bounds are

$$(0,\gamma(M))\subseteq\mathcal{S}(\mathcal{A}_M)\subseteq\mathcal{S}(\widetilde{\mathcal{A}}^u_M)\subseteq(0,\gamma(M)]$$

by [1, Corollary 4.19]. Suppose uniform surjectivity held at $\alpha=\gamma(M)$. Choose $0<s<\gamma(M)$ so that $r=\gamma(M)-s$ is positive and rational. The inverse-Borel transfer and the regularization lemma give uniform surjectivity for the strongly regular Q' on S_r, where $r=\gamma(Q')$. This contradicts the rational-opening criterion [1, Theorem 4.17]. Thus the uniform endpoint, and consequently the derivative endpoint, are excluded. □

References

[1] J. Jiménez-Garrido, J. Sanz and G. Schindl, *Injectivity and surjectivity of the asymptotic Borel map in Carleman ultraholomorphic classes*, J. Math. Anal. Appl. **469** (2019), 136–168. https://arxiv.org/abs/1805.01153; https://doi.org/10.1016/j.jmaa.2018.09.011.